# Simulation-Based Pricing and Risk Analysis for Coastal Nest Motel

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# AGENDA

1. Introduction
2. Model Description, Conceptual Model, and Assumptions
3. Decision Models
4. Scenario Analysis
5. Stochastic Variables
6. Simulated Output Distribution & Risk Analysis

# 1. Introduction

This report presents a simulation-driven decision support model developed for Coastal Nest Motel to assist management in making informed pricing and booking decisions during peak demand periods. By incorporating both deterministic and stochastic components, the model estimates daily profit under varying conditions of customer demand, cancellations, walk-ins, and operating costs. The analysis supports strategic evaluation of room pricing and cancellation fee policies, enabling the Sales Manager to assess risk, optimize revenue, and mitigate the financial impact of overbooking. Scenario and risk analysis results offer key insights to guide data-driven decision-making in a dynamic and uncertain hospitality environment.

# 2. Model Description

This model estimates Coastal Nest's daily profit by simulating revenue, operating costs, and guest behaviors under varying demand conditions. It incorporates stochastic inputs (online bookings, cancellations, walk-ins, expenses), decision variables (room price, cancellation fee), and calculated outputs to reflect real-world operations. The simulation enables the Sales Manager to assess how different input combinations influence profitability. It also supports scenario and risk analysis, helping guide data-driven pricing decisions. A conceptual model (shown next) outlines the flow of variables and calculations used.

| Stochastic inputs |
|---|
| Daily Online Bookings |
| Daily Cancellations |
| Daily Walk-ins |
| Daily Miscellaneous Expenses |

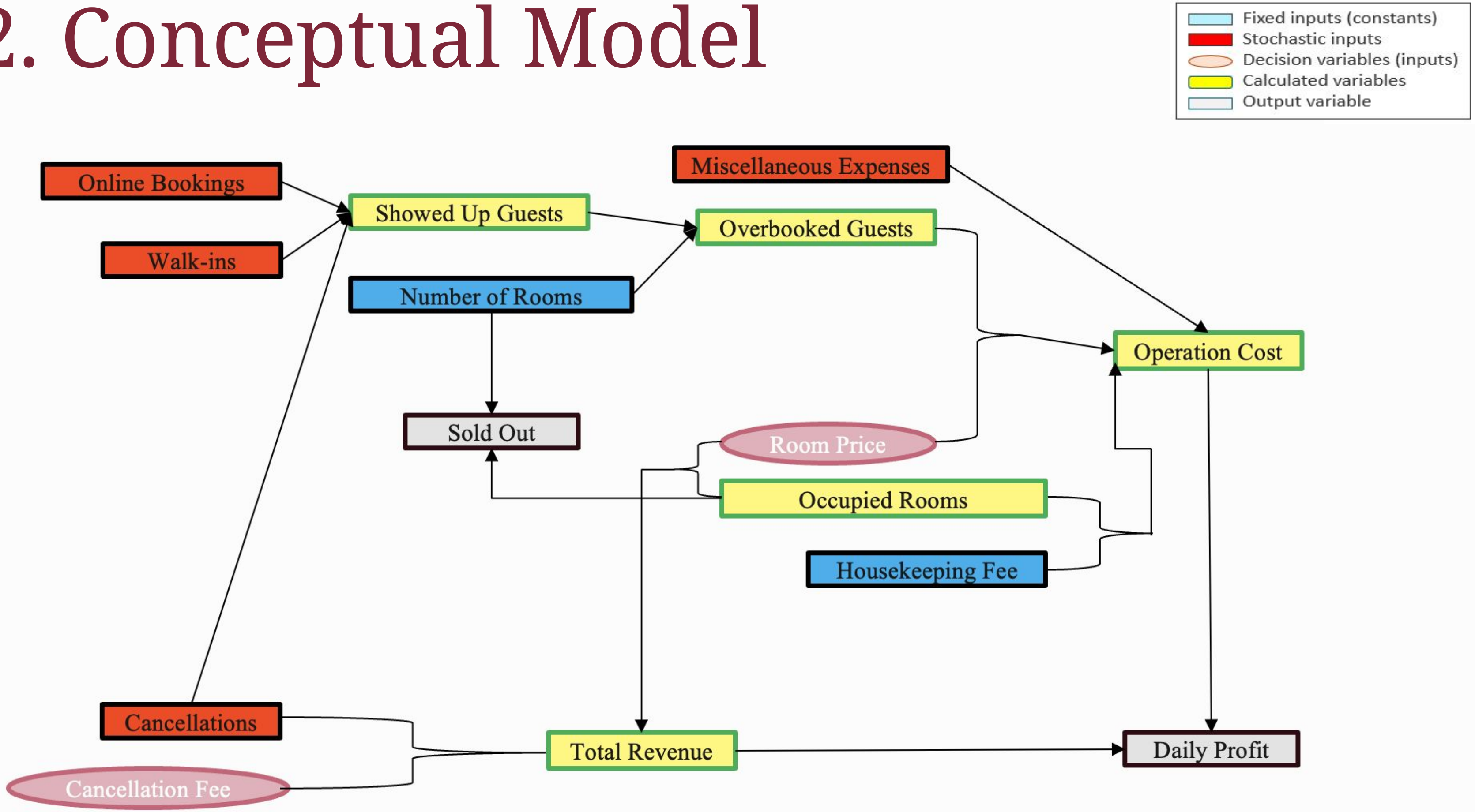
| Decision variables |
|---|
| Room Price |
| Late Cancellation Fee |

| Fixed inputs |
|---|
| Number of Rooms |
| House Keeping Cost per Room |

| CALCULATED VALUES |
|---|
| Number of Occupied Rooms |
| Showed Up Guests |
| Overbooked Guests |
| Daily Operation Cost |
| Daily Total Revenue |

| OUTPUT |
|---|
| Total Daily Profit |
| Sold Out: Yes/ No |

# 2. Conceptual Model



# 2. Assumptions

## Stochastic variables:

- Online Bookings - Normal distribution ($\mu = 22$, $\sigma = 3$ for \$150; $\mu = 20$ for \$180), reflecting daily demand variability.
- Cancellations - Binomial distribution with 10–15% probability per booking, simulating individual cancellation behavior.
- Walk-ins - Discrete Uniform distribution (2–6 for \$180; 3–7 for \$150), assuming all values are equally likely.
- Miscellaneous Expenses - Triangular distribution (min = \$100, mode = \$150, max = \$200), representing irregular but bounded costs.

## Decision variables:

- Room Price is adjustable (\$150–\$180) to explore revenue outcomes.
- Cancellation Fee ranges from \$50–\$70, used to discourage late cancellations and recover costs.

## Fixed variables:

- Number of Rooms: 25 rooms available per day.
- Housekeeping Cost: \$30 per occupied room.

# 2. Assumptions

## Calculated variables:

- The number of rooms filled by guests. Occupied Rooms = MIN(Rooms, Showed-Up Guests).
- The total number of guests who arrive. Showed-Up Guests = Bookings + Walk-ins – Cancellations.
- The total number of guests who arrive. Overbooked Guests = MAX(Showed-Up – Rooms, 0).
- Operation Cost = Housekeeping + Compensation + Misc Expenses.
- Total Revenue = Room Revenue + Cancellation Fee Revenue.

## Output:

- Profit = Total Revenue – Operation Cost.
- Sold Out = “Yes” if Occupied = 25; otherwise, “No”.

# 3. Decision Model

**CALCULATED VALUES & OUTPUT** values

| | |
|---|---|
| Number of Occupied Rooms | 21 |
| Showed Up Guests | 21 |
| Overbooked Guests | 0 |
| Daily Operation Cost | $780 |
| Daily Total Revenue | $3.930 |

**OUTPUT**

| | |
|---|---|
| Total Daily Profit | $3.150 |
| Sold Out: Yes/ No | No |

**INPUTS**

| Stochastic inputs | values |
|---|---|
| Daily Online Bookings | 20 |
| Daily Cancellations | 3 |
| Daily Walk-ins | 4 |
| Daily Miscellaneous Expenses | $150 |

**Decision variables**

| | |
|---|---|
| Room Price | $180 |
| Late Cancellation Fee | $50 |

| Fixed inputs | values |
|---|---|
| Number of Rooms | 25 |
| House Keeping Cost per Room | $30 |

- Room price is chosen as the primary decision variable due to its strong influence on daily profit and flexibility for dynamic adjustment. Unlike the relatively fixed cancellation fee, room pricing directly drives revenue and offers greater control for optimizing performance under changing demand.

# 4. Scenario Analysis – $180 Room

| Scenario Summary | | | | |
|---|---|---|---|---|
| | Current Values: | Best case | Base case | Worst case |
| **Changing Cells:** | | | | |
| daily.online.reservatio | 20 | 23 | 20 | 17 |
| **Result Cells:** | | | | |
| Total.daily.profit | $3.150 | $3.600 | $3.150 | $2.700 |

| Scenario Summary | | | | |
|---|---|---|---|---|
| | Current Values: | Best case | Base case | Worst case |
| **Changing Cells:** | | | | |
| late.cancellation | 3 | 1 | 3 | 5 |
| **Result Cells:** | | | | |
| Total.daily.profit | $3.150 | $3.350 | $3.150 | $2.950 |

## Daily Online Bookings:

- Profit is highly sensitive to bookings. Increasing from 17 to 23 bookings raises profit from $2,700 to $3,600, a $900 swing. This highlights the importance of demand generation while avoiding over-booking.

## Daily Cancellations:

- Higher cancellations reduce profit, but the impact is smaller ($400 range). Revenue from cancellation fees partly offsets the loss, but profit is still highest when cancellations are minimized (best case: $3,350 with 1 cancellation).

# 4. Scenario Analysis – $180 Room

| Scenario Summary | Current Values: | Best case | Base case | Worst case |
|---|---|---|---|---|
| **Changing Cells:** | | | | |
| walk.ins | 4 | 6 | 4 | 2 |
| **Result Cells:** | | | | |
| Total.daily.profit | $3.150 | $3.450 | $3.150 | $2.850 |

| Scenario Summary | Current Values: | Best case | Base case | Worst case |
|---|---|---|---|---|
| **Changing Cells:** | | | | |
| Miscellaneous.expens | 150 | 120 | 150 | 170 |
| **Result Cells:** | | | | |
| Total.daily.profit | $3.150 | $3.180 | $3.150 | $3.130 |

## Daily Walk-ins:

- More walk-ins improve profit moderately (from $2,850 to $3,450), but not as significantly as online bookings. This suggests walk-ins are a useful supplement but not a core revenue driver.

## Miscellaneous Expenses:

- Changes in daily expenses have the least impact. Even at the highest cost ($170), profit only drops by $50. This confirms the model is revenue-sensitive more than cost-sensitive.

# 4. Scenario Analysis – $150 Room

| Scenario Summary | | | | |
|---|---|---|---|---|
| | Current Values: | Best case | Base case | Worst case |
| Changing Cells: | | | | |
| daily.online.reservatio | 22 | 25 | 22 | 19 |
| Result Cells: | | | | |
| Total.daily.profit | $2.810 | $2.900 | $2.810 | $2.450 |

| Scenario Summary | | | | |
|---|---|---|---|---|
| | Current Values: | Best case | Base case | Worst case |
| Changing Cells: | | | | |
| late.cancellation | 4 | 2 | 4 | 6 |
| Result Cells: | | | | |
| Total.daily.profit | $2.810 | $2.950 | $2.810 | $2.670 |

## Daily Online Bookings:

- The profit range of $2,450 to $2,900 indicates a moderate level of revenue potential under the $150 room pricing strategy. In contrast to the $180 room scenario, increasing bookings beyond capacity in this setting leads to overbooking compensation, which reduces profit.

## Daily Cancellations:

- The profit range of $2,450 to $2,900 reflects moderate revenue potential under the $150 room pricing scenario. An increase in daily cancellations reduces the number of guests arriving, which diminishes overall revenue. Although cancellation fees offer some financial offset, the lower room rate limits their impact.

# 4. Scenario Analysis – $150 Room

| Scenario Summary | Current Values: | Best case | Base case | Worst case |
|---|---|---|---|---|
| Changing Cells: | | | | |
| walk.ins | 5 | 7 | 5 | 3 |
| Result Cells: | | | | |
| Total.daily.profit | $2.810 | $3.050 | $2.810 | $2.570 |

| Scenario Summary | Current Values: | Best case | Base case | Worst case |
|---|---|---|---|---|
| Changing Cells: | | | | |
| Miscellaneous.expens | 150 | 120 | 150 | 170 |
| Result Cells: | | | | |
| Total.daily.profit | $2.810 | $2.840 | $2.810 | $2.790 |

## Daily Walk-ins:

- Profit increases modestly from $2,570 to $3,050 as walk-ins rise. Walk-ins again help stabilize revenue but do not drastically shift profitability.

## Miscellaneous Expenses:

- These show minimal influence. Profit only shifts by $50 across all three levels, reinforcing that managing bookings and pricing is more critical than controlling small cost fluctuations.

# 4. Scenario Analysis

1) Daily profit is highly sensitive to online bookings. At $180 pricing, profit varies between $2,700 and $3,600 (+$900). At $150 pricing, it ranges from $2,450 to $2,900 (+$450).

➤ Maximizing online bookings while avoiding overbooking is critical for revenue optimization.

2) Cancellations reduce earnings: Profit drops by $400 at $180 and $280 at $150 as cancellations rise.

➤ Fees recover some loss but not all missed room revenue.

3). Walk-ins help but less so: They boost profit up to $3,450 ($180) and $3,050 ($150), though less reliably.

➤ Volatility and overbooking penalties reduce their reliability as a primary revenue source.

4). Expenses matter least: Profit shifts by only $50–$90 across cost scenarios.

➤ Demand-side shifts matter more than cost variability.

5) Lower pricing reduces financial resilience. At $150, profits are consistently lower, even in best-case demand scenarios.

➤ Tighter margins limit the ability to absorb shocks from cancellations or overbooking.

6) Peak profit is achieved with high demand balanced within capacity.

➤ This reinforces that accurate forecasting is more effective than aggressive overbooking.

| Scenarios | Best | Base | Worst |
|---|---|---|---|
| $180 Room with $50 Cancellation Fee | | | |
| daily.online.reservation | 23 | 20 | 17 |
| Total.daily.profit | $3.600 | $3.150 | $2.700 |
| late.cancellation | 1 | 3 | 5 |
| Total.daily.profit | $3.350 | $3.150 | $2.950 |
| walk.ins | 6 | 4 | 2 |
| Total.daily.profit | $3.450 | $3.150 | $2.850 |
| Miscellaneous.expenses | 120 | 150 | 170 |
| Total.daily.profit | $3.180 | $3.150 | $3.130 |
| $150 Room with $50 Cancellation Fee | | | |
| daily.online.reservation | 25 | 22 | 19 |
| Total.daily.profit | $2.900 | $2.810 | $2.450 |
| late.cancellation | 2 | 4 | 6 |
| Total.daily.profit | $2.950 | $2.810 | $2.670 |
| walk.ins | 7 | 5 | 3 |
| Total.daily.profit | $3.050 | $2.810 | $2.570 |
| Miscellaneous.expenses | 120 | 150 | 170 |
| Total.daily.profit | $2.840 | $2.810 | $2.790 |

# 5. Stochastic Variables

The model includes four stochastic variables:

- Daily online bookings
- Daily cancellations
- Daily walk-ins
- Miscellaneous expenses

Each variable was assigned a probability distribution to reflect its real-world uncertainty:

- Online Bookings – Normal Distribution: Bookings fluctuate around a central mean due to factors like weekday patterns, pricing, and seasonality. The symmetric shape of the normal distribution effectively captures this variability.
- Cancellations – Binomial Distribution: Each online reservation is treated as an independent trial with a fixed probability of being cancelled. This structure makes the binomial distribution ideal for modelling binary outcomes.
- Walk-ins – Discrete Uniform Distribution: Since there is no clear trend or preference in walk-in numbers, all values within a defined range are considered equally likely. This reflects random walk-in behavior.
- Miscellaneous Expenses – Triangular Distribution: Based on expert estimates of minimum, most likely, and maximum values to reflect skewed cost variability.

| **Stochastic inputs** |
|---|
| **Daily Online Bookings** |
| **Daily Cancellations** |
| **Daily Walk-ins** |
| **Daily Miscellaneous Expenses** |

# 6. Simulated Output

The simulation generated 1,000 iterations for each pricing and cancellation-fee combination, capturing a robust range of profit scenarios under uncertainty.

**$180 Room - $50 Late Cancellation Fee:** Mean Profit: $3,137.36 | Min: $1,720 | Max: $3,824 | Standard Deviation: $384.52

- The profit distribution is slightly left-skewed, indicating a higher frequency of high-profit days, with lower profits occasionally caused by overbooking compensation or cancellations.

**$180 Room - $70 Late Cancellation Fee:** Mean Profit: $3,157.55 | Min: $1,758 | Max: $3,880 | Standard Deviation: $376.13

- Raising the fee slightly improved the mean while slightly reducing variability. It suggests that pricing cancellations more aggressively can modestly buffer against demand shocks.

**$150 Room with $50 Late Cancellation Fee:** Mean Profit: $2,624.06 | Min: $1,430 | Max: $3,157 | Standard Deviation: $293.10

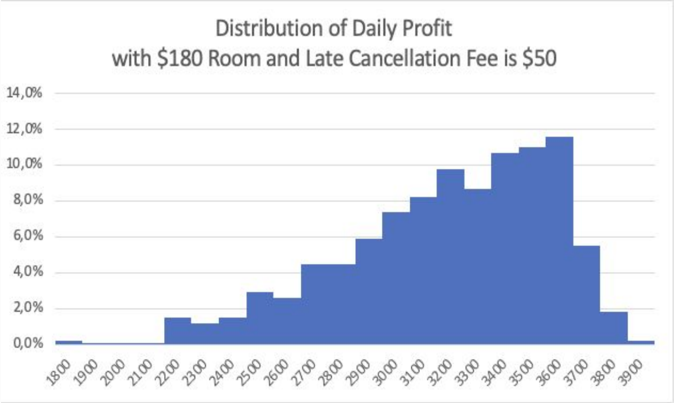
- A lower room rate reduces mean profit and increases risk of low-profit days, as seen in the lower minimum and higher skewness.

**$150 Room with $70 Late Cancellation Fee:** Mean Profit: $2,685.79 | Min: $1,567 | Max: $3,385 | Standard Deviation: $297.25

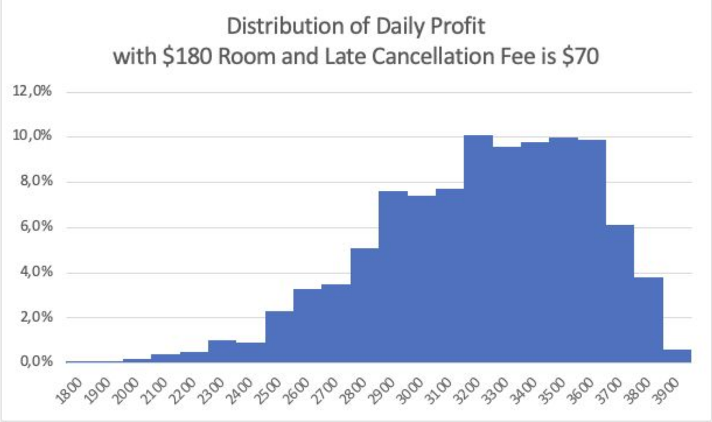
- Profit improved modestly over the $50 fee version.

- Across all outputs, the $180 pricing outperforms $150 in both profit and resilience. Meanwhile, increasing the cancellation fee tends to improve average returns and reduce risk, though with diminishing gains.

# 6. Simulated Output

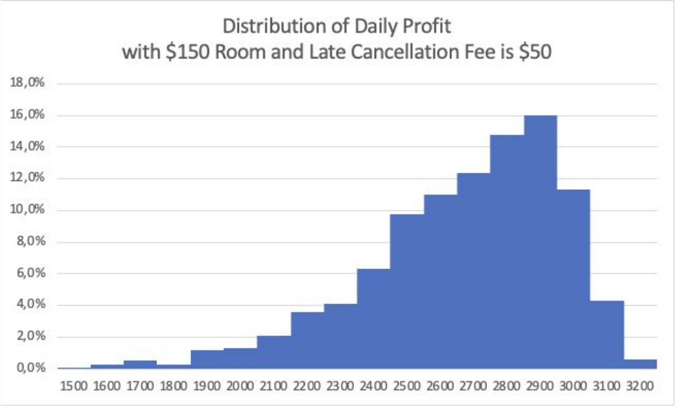



- Histogram of $180 Room and $50 cancellation fee shows a right-shifted peak between $3,300 and $3,600.
- Distribution is slightly left-skewed, with a long left tail due to occasional low-profit days (e.g., high cancellations or low bookings).
- Most likely profit lies around $3,400–$3,600, confirming the descriptive mode range.



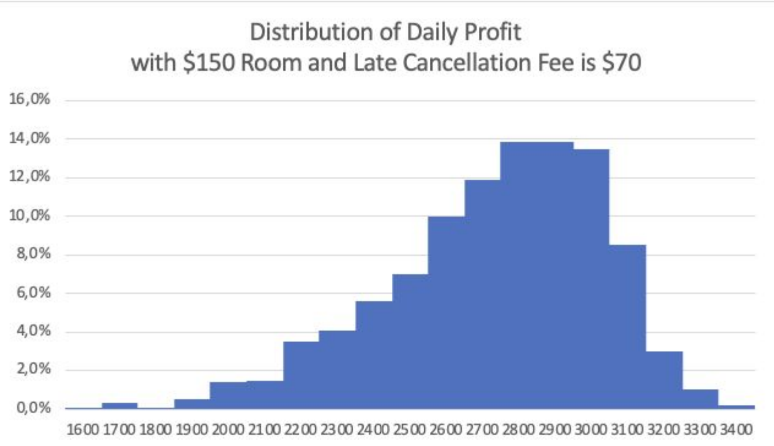

- Histogram of $180 Room and $70 cancellation fee is narrower and taller, peaking sharply between $3,200–$3,600.
- Suggests higher profit concentration and reduced variance, indicating more consistent performance.

# 6. Simulated Output




- The histogram of $150 room with $50 cancellation fee is wider and flatter, peaking around $2,700–$2,900.
- Indicates more uncertainty and a higher likelihood of suboptimal profit.

- Histogram of $150 room with $70 cancellation fee shows slight improvement in concentration; most values fall between $2,700–$3,000.
- However, low-profit outcomes below $2,000 are still present.



# 6. Risk Analysis

**RISK PROFILING** — **$180 Room and $50 Late Cancellation Fee**

| | $1500 to $2000 | $2000 to $2500 | $2500 to $3000 | $3000 to $3500 | Over $3500 |
|---|---|---|---|---|---|
| Probability | 0,4% | 7,2% | 24,9% | 48,4% | 19,1% |
| Daily Online Bookings | 14 | 28 | 28 | 28 | 28 |

- This scenario delivers balanced profitability and controlled risk.
- Only 7.6% of outcomes fall below $2,500, while 73.3% stay in the $2,500–$3,500 range.
- 19.1% of simulations exceed $3,500, indicating solid upside potential.
- Best suited when demand forecasting is reliable, minimizing overbooking

**RISK PROFILING** — **$180 Room and $70 Late Cancellation Fee**

| | $1500 to $2000 | $2000 to $2500 | $2500 to $3000 | $3000 to $3500 | Over $3500 |
|---|---|---|---|---|---|
| Probability | 0,4% | 5,1% | 26,9% | 47,2% | 20,4% |
| Daily Online Bookings | 15 | 26 | 28 | 28 | 27 |

- Risk distribution is very similar to the $50 fee, but with slightly more high-end returns.
- Just 5.5% fall below $2,500, and 74.1% remain in the profitable middle band.
- 20.4% of outcomes surpass $3,500 — the highest top-tier probability among all cases.
- Enhances revenue from cancellations without harming stability.

# 6. Risk Analysis

**RISK PROFILING** — **$150 Room and $50 Late Cancellation Fee**

| | $1000 to $1500 | $1500 to $2000 | $2000 to $2500 | $2500 to $3000 | Over $3000 |
|---|---|---|---|---|---|
| Probability | 0,1% | 3,6% | 25,9% | 65,5% | 4,9% |
| Daily Online Bookings | 11 | 31 | 30 | 30 | 28 |

- Most conservative profile: low revenue ceiling but stable outcomes.
- 91.4% of simulations are below $3,000, with 65.5% concentrated in the $2,500–$3,000 range.
- Only 4.9% exceed $3,000, showing limited upside.
- Suited for minimizing risk, not maximizing profit.

**RISK PROFILING** — **$150 Room and $70 Late Cancellation Fee**

| | $1500 to $2000 | $2000 to $2500 | $2500 to $3000 | Over $3000 |
|---|---|---|---|---|
| Probability | 2,4% | 21,7% | 63,2% | 12,7% |
| Daily Online Bookings | 31 | 31 | 29 | 29 |

- Slightly better than the $50 fee version, but still constrained upside.
- 12.7% exceed $3,000, while 63.2% stay between $2,500–$3,000.
- Risk below $2,000 is low (2.4%), suggesting moderate resilience.

# 6. Risk Analysis - Summary

- The $180 room with a $70 cancellation fee is the most effective strategy, delivering the highest average profit, 20.4% chance of exceeding $3,500, and only 0.4% risk of falling below $2,000. It balances revenue and operational consistency well. The $180 room with a $50 fee is also effective, offering similar mid-range stability and slightly lower upside (19.1%).

- In contrast, $150 room options underperform in both profit and risk. Even with a $70 fee, upside remains limited (12.7% > $3,000), and the $150 with $50 fee shows the weakest results—lowest returns and highest low-profit risk.

# 7. Recommendation & Conclusion

Adopt $180 Room with $70 Fee

- Delivers the best results, highest average profit, strong upside (20.4% > $3,500), and minimal downside. Ideal for peak periods with predictable demand.

Limit Overbooking to 2–3 Rooms

- Excessive overbooking cuts into profits due to compensation costs. Only exceed this range with strong historical cancellation support.

Avoid $150 Room with $50 Fee

- Lowest performance and highest risk. Not recommended for high-demand periods.

Use $150 Room with $70 Fee for Off-Peak

- Offers better consistency than its $50 counterpart. Use when stability is prioritized over profit.

Monitor Trends and Adjust Policies

- The model's performance depends on the alignment between forecasted cancellations and real demand. Management should review historical booking and cancellation trends frequently and adjust overbooking or cancellation fee policies accordingly.

- This project applied a simulation-based decision model to evaluate pricing strategies for Coastal Nest Motel under uncertainty. Through scenario and risk analysis, it was found that room price has the greatest impact on profitability, while the cancellation fee helps mitigate revenue loss from no-shows. The $180 room price with a $70 late cancellation fee delivers the best balance of high returns and low risk, making it the recommended strategy for peak periods. The model supports data-driven decision-making and enables the Sales Manager to respond confidently to fluctuating demand and operational variability.

# THANK YOU!